Math Inquirers

Extension Workbook

with answers

Level 3

Self-study | Ages 8-10



My name _____

About this book

Math Inquirers Extension Workbook, Level 3
Self-study practice with answers.

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- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
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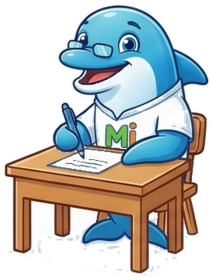
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Edition 1, 2026. First printing.

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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Number sleuths (numbers to 10 000)	Not in this proof edition				☐
2	Cryptarithm crackers	Not in this proof edition				☐
3	Times table treasure	Not in this proof edition				☐
4	Model makers (bar models)	Not in this proof edition				☐
5	Remainder riddles	Not in this proof edition				☐
6	Fraction walls	Not in this proof edition				☐
7	Area architects	Not in this proof edition				☐
8	Angle and line detectives	Not in this proof edition				☐
9	Measure and money masters	Not in this proof edition				☐
10	Puzzle day	6-7	8-9	10 and 12	11	☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

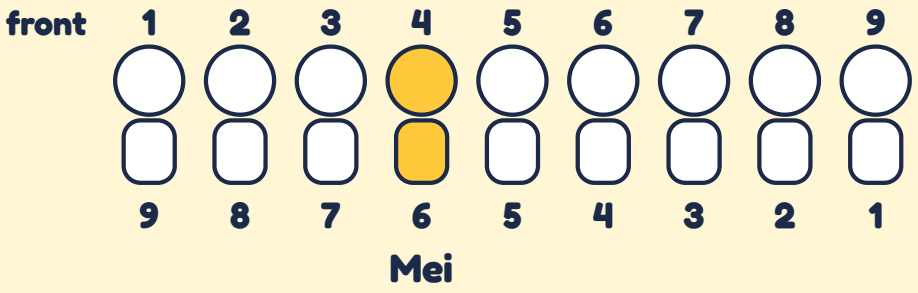
Learn it: Mixed puzzles

My goal. I can start a puzzle by drawing it, listing it or making a table. I can read a bar graph and solve logic puzzles.

To start a puzzle, try one of 3 ways. Draw it, list it, or make a table. Then check every clue.

Worked example

Mei is 4th from the front and 6th from the back of a line. How many children are in the line?



1. **Draw the line. Put Mei in 4th place from the front.**
2. **Count from Mei to the back: Mei is 6th from the back, so 5 children are behind her.**
3. **3 children are in front of Mei, then Mei, then 5 behind: 3 + 1 + 5 = 9.**

Answer: There are 9 children in the line.

Three ways to start

1. Draw it: make a picture of the puzzle.
2. List it: write every case in order.
3. Table: write what you know in rows and columns.

Your turn

Answer at the back of the book.

Try **Siti is 5th from the front and 3rd from the back. How many children are in the line?** ☐

Start here

Do these first. Take your time.

1 **The bar graph shows the books 4 children read. Kumar read 15 more books than Siti. Draw Kumar's bar on the graph.**

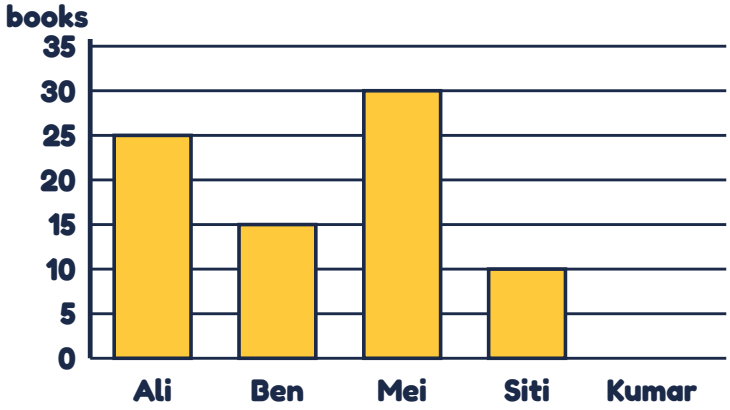
(a) Who read the most books?

Ali Ben Mei Siti

(b) Mei read ☐ more books than Ben.

(c) Kumar read ☐ books.

(d) The 5 children read ☐ books in all.



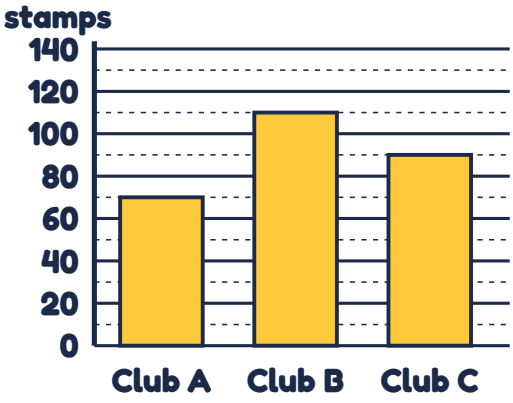
2 **The graph shows the stamps 3 clubs collected. Each thin line is 10 stamps.**

(a) Club A collected ☐ stamps.

(b) Club B collected ☐ stamps.

(c) Club C collected ☐ stamps.

(d) All 3 clubs collected ☐ stamps.



Start here

Keep going. Check your work.

4 **Draw it. Count the children in each line.**

(a) Ravi is 3rd from the front. Mei is last. 5 children stand between them. There are ☐ children in the line.

(b) Aisha is 7th from the front and 3rd from the back. There are ☐ children in the line.

Draw (a):

Draw (b):

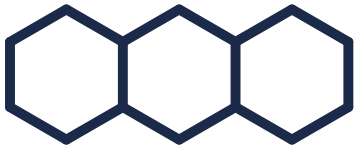
5 **Use the cards 3, 6 and 9 once each. List ALL the 3-digit numbers you can make. Begin with the smallest.**

(a) _____

(b) There are ☐ even numbers in your list.

(c) The number closest to 700 is ☐.

6 **Hexagons are joined in a row with sticks. 1 hexagon needs 6 sticks. 2 hexagons need 11 sticks. 3 hexagons need 16 sticks. Make a table.**



(a) 6 hexagons need ☐ sticks.

(b) 41 sticks make ☐ hexagons.

hexagons	sticks

Start here

Nearly there. You can do it.

8 Two numbers add up to 60. One number is 14 more than the other. What are the numbers? ☐ and ☐

Guess, check, improve: _____

9 A snail climbs a wall 10 m high. Each day it climbs up 4 m. Each night it slides down 1 m. On which day does it reach the top? Day ☐

day	in the morning	at night

10 The 3rd of a month is a Tuesday.

(a) The 24th is a Mon Tue Wed Thu Fri Sat Sun

(b) The 30th is a Mon Tue Wed Thu Fri Sat Sun

11 Aisha has $10. A juice costs $1.85 and a sandwich costs $3.40. She buys 2 juices and 1 sandwich. Her change is $ ☐.

12 I think of a number. I double it and add 5. The answer is 21. My number is ☐.

13 Look for a pattern. 2, 7, 12, 17, … The 12th number is ☐.

Working: _____

Think harder

Hints are upside down. Try first!

1 **7 friends meet. Each friend shakes hands once with every other friend. How many handshakes are there?** ☐

friends	handshakes

Hint. Try 2, 3, 4 friends first. Make a table.

2 **One of Ben, Aisha and Wei broke a cup. Ben says, "Wei did it." Aisha says, "I did not do it." Wei says, "Ben is telling a lie." Only ONE of them is telling the truth. Who broke the cup? Ben Aisha Wei**

If Ben did it: _____

If Aisha did it: _____

If Wei did it: _____

Hint. Test each child as the one who broke the cup.

3 **A graph shows the pencils sold on 4 days. Tuesday sold 15 more than Monday. Wednesday sold twice as many as Monday. Thursday sold 10 fewer than Wednesday. They sold 185 pencils in all.**

(a) Monday: ☐ (b) Tuesday: ☐ (c) Wednesday: ☐

(d) Thursday: ☐

Draw your bars:

_____

Hint. Draw Monday as 1 bar. Then draw the other days with it.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **How many squares of any size can you find in a 4 by 4 grid of squares? Start with the 1 by 1 squares.**

(a) 1 by 1 squares: ☐ (b) 2 by 2 squares: ☐

(c) 3 by 3 squares: ☐ (d) All the squares: ☐

Hint. A 2 by 2 square can sit in 3 places across and 3 down.

2 **Beyond Level 3: A balance puzzle.**
9 coins look the same. One coin is lighter than the others. You have a balance scale.

(a) With 3 coins, you can always find the light coin in ☐ weighing.

(b) With 9 coins, you can always find it in ☐ weighings.

Hint. Put 3 coins on each side. What does the scale tell you?

3 **Ali has 8 coins. They are 10¢ coins and 20¢ coins. They are worth $1.50 in all. How many of each coin does he have?**

(a) ☐ ten-cent coins

(b) ☐ twenty-cent coins

10¢ coins	20¢ coins	value

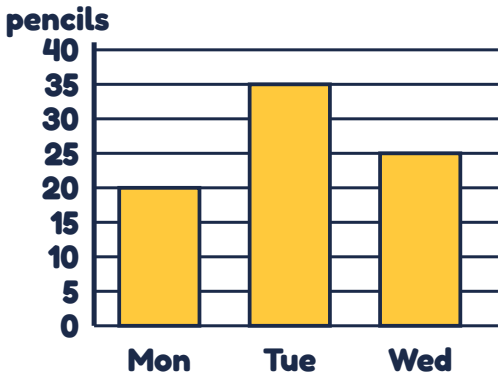
Hint. Try 4 of each. How much are they worth?

✔ Check yourself

No hints. Try each one on your own.

1

The graph shows the pencils sold on 3 days. Tuesday sold ☐ more pencils than Monday.



2 Mei is 5th from the front and 4th from the back of a line. There are ☐ children in the line.

Working: _____

3 Use the cards 1, 4 and 7 once each to make 3-digit numbers. ☐ of the numbers are even.

Working: _____

4 Look for a pattern. 2, 6, 10, 14, … The 8th number is ☐.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number sleuths (numbers to 10 000)	☐	☐	☐	☐	/ 4	
2	Cryptarithm crackers	☐	☐	☐	☐	/ 4	
3	Times table treasure	☐	☐	☐	☐	/ 4	
4	Model makers (bar models)	☐	☐	☐	☐	/ 4	
5	Remainder riddles	☐	☐	☐	☐	/ 4	
6	Fraction walls	☐	☐	☐	☐	/ 4	
7	Area architects	☐	☐	☐	☐	/ 4	
8	Angle and line detectives	☐	☐	☐	☐	/ 4	
9	Measure and money masters	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 3.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 10 · Puzzle day

Your turn
page 6

Answer: 7
Why: 4 children are in front of Siti and 2 are behind her. 4 + 1 + 2 = 7. Check: 5 + 3 - 1 = 7.

START HERE

1
page 7

Answer: (a) Mei **(b)** 15 **(c)** 25 **(d)** 105
Why: (a) The tallest bar is Mei's. (b) 30 - 15 = 15. (c) Siti read 10 books, so Kumar read 10 + 15 = 25. (d) 25 + 15 + 30 + 10 + 25 = 105.

2
page 7

Answer: (a) 70 **(b)** 110 **(c)** 90 **(d)** 270
Why: The labelled lines go up in 20s and there is a thin line halfway between them, so each thin line is 10. A is at 70, B is at 110 and C is at 90. Together: 70 + 110 + 90 = 270.

3
page 7

Answer: (a) grapes **(b)** apple **(c)** banana **(d)** orange
Why: Only Lily has a name with 4 letters, so Lily has the orange. Kumar has neither the apple nor the banana, and the orange is taken, so Kumar has the grapes. Wei does not have the apple, so Wei has the banana. Aisha has the apple.

4
page 8

Answer: (a) 9 **(b)** 9
Why: (a) Ravi is 3rd. The 5 children between are 4th to 8th. Mei is 9th and last. (b) 6 children are in front of Aisha and 2 are behind her: 6 + 1 + 2 = 9.

5
page 8

Answer: (a) 369, 396, 639, 693, 936, 963 **(b)** 2 **(c)** 693
Why: Fix the first digit and list the rest in order: 369, 396, then 639, 693, then 936, 963. The even numbers end in 6: 396 and 936. 693 is only 7 from 700. The others are much farther away.

6
page 8

Answer: (a) 31 **(b)** 8
Why: Each new hexagon adds 5 sticks. The table goes 1: 6, 2: 11, 3: 16, 4: 21, 5: 26, 6: 31. For 41 sticks, count on: 7 hexagons need 36 and 8 hexagons need 41.

7
page 8

Answer: 162, 252 and 342
Why: The hundreds digit is 1, 2 or 3. The ones digit is even and not 0, so it is 2, 4 or 6, and the tens digit is greater. With 1: tens + ones = 8, so 6 and 2. With 2: tens + ones = 7, so 5 and 2. With 3: tens + ones = 6, so 4 and 2. The numbers are 162, 252 and 342.

8
page 9

Answer: 23 and 37
Why: Take away the extra 14: 60 - 14 = 46. Share it into 2 equal parts: 23 each. The numbers are 23 and 23 + 14 = 37. Check: 23 + 37 = 60.

9
page 9

Answer: 3
Why: Day 1: it climbs to 4 m and slides back to 3 m. Day 2: it climbs to 7 m and slides back to 6 m. Day 3: it climbs from 6 m to 10 m. It reaches the top on day 3 and does not slide back.

10 page 9
Answer: (a) Tue (b) Mon
Why: (a) From the 3rd to the 24th is 21 days, which is 3 weeks. So the 24th is a Tuesday again. (b) From the 3rd to the 30th is 27 days = 3 weeks and 6 days. 6 days after Tuesday is Monday.

11 page 9
Answer: 2.90
Why: 2 juices cost 2 × 1.85 = 3.70. With the sandwich: 3.70 + 3.40 = 7.10. Change: 10.00 - 7.10 = 2.90.

12 page 9
Answer: 8
Why: Work backwards. Take away 5 from 21: 16. Then halve it: 8. Check: 8 × 2 = 16, and 16 + 5 = 21.

13 page 9
Answer: 57
Why: The numbers go up by 5 each time. The 12th number is the 1st number plus 11 jumps: 2 + 11 × 5 = 57.

THINK HARDER

1 page 10
Answer: 21
Why: With 2 friends: 1 handshake. With 3: 3. With 4: 6. Each new friend shakes hands with all the friends already there: 5 friends add 4 (10), 6 add 5 (15) and 7 add 6 (21).

2 page 10
Answer: Aisha
Why: If Ben did it: Aisha tells the truth, and Wei tells the truth (Ben's words are a lie). That is 2 true statements. If Wei did it: Ben and Aisha both tell the truth. If Aisha did it: Ben lies, Aisha lies and Wei tells the truth. That is exactly 1. So Aisha did it.

3 page 10
Answer: (a) 30 (b) 45 (c) 60 (d) 50
Why: Monday is 1 box. Tuesday is 1 box and 15. Wednesday is 2 boxes. Thursday is 2 boxes take away 10. In all: 6 boxes + 15 - 10 = 6 boxes + 5 = 185. So 6 boxes = 180 and 1 box = 30. Monday 30, Tuesday 45, Wednesday 60, Thursday 50. Check: 30 + 45 + 60 + 50 = 185.

GO FURTHER

1 page 11
Answer: (a) 16 (b) 9 (c) 4 (d) 30
Why: There are 16 small squares. A 2 by 2 square fits in 3 places across and 3 down: 9. A 3 by 3 square fits in 2 places across and 2 down: 4. There is 1 big square. 16 + 9 + 4 + 1 = 30.

2 page 11
Answer: (a) 1 (b) 2
Why: (a) Put 1 coin on each side. If the scale tips, the light coin is on the high side. If it balances, it is the coin left out. That is 1 weighing. (b) Weigh 3 coins against 3 coins. This shows which group of 3 has the light coin. Then use the 1 weighing from (a). That is 2 weighings.

3
page 11

Answer: (a) 1 (b) 7
Why: If all 8 coins were 10¢, they would be worth 80¢. They are worth 150¢, which is 70¢ more. Changing one 10¢ coin to a 20¢ coin adds 10¢, so 7 coins are 20¢ and 1 coin is 10¢. Check: 10 + 7 × 20 = 150.

4
page 11

Answer: Answers vary. Example: Lily, Wei and Kumar each play a different game: chess, ball or cards. Lily does not play ball. Wei plays neither chess nor cards. Kumar does not play chess. Answer: Lily plays chess, Wei plays ball, Kumar plays cards.
Why: Check that your clues are true for your answer and that every other arrangement breaks a clue. In the example, Wei plays ball (clue 2), then Kumar cannot play chess (clue 3), so Kumar plays cards and Lily plays chess.

CHECK YOURSELF

1
page 12

Answer: 15
Why: Monday is at 20 and Tuesday is at 35. 35 - 20 = 15.

2
page 12

Answer: 8
Why: 4 children are in front of Mei and 3 are behind her. 4 + 1 + 3 = 8.

3
page 12

Answer: 2
Why: The numbers are 147, 174, 417, 471, 714 and 741. The even ones end in 4: 174 and 714.

4
page 12

Answer: 30
Why: The numbers go up by 4. The 8th number is 2 + 7 × 4 = 30.